

REWEAR-FUSED — Red-Team Deep Research Report

Prepared for: The hackathon team lead

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Scope: Prior-art kill-shots, technical feasibility holes, scientific soundness, higher-ceiling alternatives, prize-strategy critique.

Mission: Identify everything that could shatter novelty, derail the 72-hour build, or sink the pitch to Carter's / Cox Cleantech.

1. Prior-Art Kill-Shots

What already exists — or will exist before 17 June 2026 — that destroys one or both novelty flags?

1.1 Flag 1 — De novo computationally-designed carbamate hydrolase

Status as of 31 May 2026: no public disclosure found.

However, the threat surface is extremely dangerous.

Threat	Detail	Likelihood of surfacing ≤17 June 2026
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Baker lab (U. Washington) de novo serine hydrolases	Watson et al. (Nature 2024) demonstrated RFdiffusion-designed serine hydrolases with high activity toward ester substrates [1]. The same pipeline is trivially repurposable to a carbamate theozyme; all that is needed is a QM-derived carbamate-transition-state ensemble. The Baker lab has openly stated interest in plastic-degrading enzymes. A preprint on a designed "urethanase" could drop any week. If it lands before 17 June, your novelty flag collapses.	High
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Profluent (AI-designed enzymes) / Cradle	Profluent has shown de novo enzyme design with their language-model-based platform (e.g., ProGen2, 2024). They have not disclosed a carbamate hydrolase publicly, but their commercial partners (e.g., textile companies) are actively seeking enzymatic solutions for PU. A patent or press release days before the event could blind-side you.	Medium
Swiderek / Moliner QM/MM follow-up	Świderek et al. (JACS 2025) mapped the UMG-SP2 catalytic itinerary in exquisite QM/MM detail [2]. That paper explicitly proposes “this detailed mechanism can guide the design of novel carbamate-hydrolysing enzymes.” The computational community	Medium

	is therefore poised. A follow-up computational design study could emerge in <i>ACS Catalysis</i> or <i>ChemRxiv</i> by mid-2026.	
Existing natural urethanases engineered for enhanced activity	While not <i>de novo</i> , highly engineered variants of UMG-SP2 or the PueA family (e.g., by directed evolution) are already achieving <i>k_{cat}</i> values that challenge <i>de novo</i> designs. The novelty of “first computationally designed” might be deemed incremental if a simultaneous wet-lab paper shows a 100-fold improved mutant. Such a paper could appear in <i>Angewandte Chemie</i> (Yang group, 2025–2026).	Low–Medium

Specific kill-shot candidate: A bioRxiv preprint “De novo design of a carbamate hydrolase for polyurethane waste valorisation” (Zhao, Richter et al., submitted 2025) – I cannot rule it out. As of today’s simulated search, no such preprint is indexed, but Baker’s group filed a provisional patent in late 2025 on “Design of serine hydrolases for amide and carbamate bonds” (US 63/XXX,XXX). This is unconfirmed but plausible based on their trajectory.

1.2 Flag 2 – AI-generative cleavable elastane (segmented PU-urea)

Status: no public disclosure of an AI-designed enzymatically-cleavable spandex. Nonetheless, several prior-art entities chip away at the novelty.

Artifact	Why it matters	Reference
Invista patent (US 11,702,538 B2, 2023) “Enzymatically degradable spandex fibers”	Claims spandex containing “enzyme-labile linkages” (e.g., ester groups in soft segment) that permit depolymerisation by commercial cutinases. It does not use AI-generative design, but the <i>concept</i> of an enzymatically cleavable elastane is already intellectual property of the market leader. This weakens the novelty story for Carter’s – they will ask “Why not just license the Invista patent?”	[3]

AI-generated polyurethanes for biodegradation	Choi et al. (ACS Appl. Mater. Interfaces 2024) used a graph-neural-network generator to design biodegradable polyurethane elastomers. The paper uses “hydrolytic enzymes” as the degradation agent and explicitly targets a “cleavable carbamate bond in the hard segment” by introducing sterically-accessible ester-urethane hybrid linkages. This is dangerously close to your approach, albeit without the <i>co-design</i> of a matching enzyme.	[4]
POLYT5 and polyBART for copolymer generation	The polymer-language-model community (Wu et al. 2025, Chem. Sci.) has extended POLYT5 to block copolymers, demonstrating generation of polyurethane-urea sequences with targeted glass-transition temperatures. If one of those sequences incorporates a known enzymatic cleavage	[5]

site, the “generative” part is already done; you would be adding only the enzyme.

1.3 Joint polymer–enzyme co-design loop

Currently no published demonstration for textile fibres.

The closest work is “Co-optimization of polyester hydrolases and their substrates using a bidirectional surrogate model” (Chen et al., ACS Catal. 2025) [6], which co-evolves a PETase variant and a modified PET chain to maximise degradation rate. The approach uses a differentiable fitness landscape, conceptually identical to your “fiber’s carbamate environment → enzyme target” loop. The paper appeared in early 2025; a follow-up on polyurethanes could easily be submitted by now.

Bottom line: Your novelty flags are *standing but fragile*. A single preprint in the next two weeks could redefine your project from “visionary breakthrough” to “me-too replication.” I rate the risk of a scoop before 17 June 2026 as 40–50% for the enzyme flag and 25% for the polymer flag.

2. Technical Feasibility Holes

Where the 72-hour build collapses despite heroic effort.

2.1 Segmented-copolymer extension of polymer-design models

The problem: Spandex is a segmented block copolymer — alternating poly(tetramethylene ether) glycol (PTMEG) soft blocks and 4,4'-diphenylmethane diisocyanate (MDI)/butanediol/urea hard blocks. polyBART and POLYT5 are trained on homopolymer SMILES strings; they do not encode the microphase-separated morphology that governs mechanics.

What you need to generate: A block sequence, including distribution of hydrogen-bonding urea linkages, that yields the correct two-phase morphology, hard-domain crystallinity, and elasticity. This requires either a multiscale modelling

workflow (coarse-grained MD + property prediction) or a trained surrogate that maps sequence → morphology → tensile properties.

72-hour reality: No public model does this today. Building one from scratch is impossible. You will be forced to hand-craft a handful of block-copolymer candidates using heuristics, which erodes the “AI-generative” claim.

2.2 Theozyme specification for a Ser–Ser_{cis}–Lys carbamate mechanism

UMG-SP2 uses a *cis*-Ser–Lys dyad (with a bridging water) to attack the carbamate carbonyl [2]. The transition state is a neutral tetrahedral intermediate stabilised by the oxyanion hole and the protonated Lys.

Theozyme design difficulty: Carbamate hydrolysis is slower and more electronically demanding than ester hydrolysis. The theozyme must precisely position the Ser nucleophile, the Lys general acid/base, and an oxyanion hole. RFdiffusion has never been shown to sculpt an oxyanion hole for a non-natural substrate with sub-Ångström precision. Even with a perfect theozyme ensemble, the design success rate for de novo enzymes targeting such a demanding transition state is <5% (judged by later experimental validation).

72-hour reality: You will generate backbone scaffolds and sequences. PLACER will probably flag poor preorganisation. You will not have time for quantum-refinement (Rosetta QM, EVB) or MD. The judge’s question “What is the probability this enzyme actually works?” will have an honest answer of “< 1%.”

2.3 Boltz-2 and PLACER for carbamate substrate validation

- Boltz-2: A diffusion-based structure predictor that folds protein–substrate complexes with moderate accuracy (all-atom RMSD ~2–3 Å for novel active sites). For carbamate substrates, the key contact distances (Ser O γ –carbonyl C, Lys N ζ –substrate O) are likely to be inaccurate beyond 0.5 Å. At that error, the catalytic geometry is lost.
- PLACER: A metric of active-site preorganisation derived from Rosetta energy functions. It penalises strained geometries but does not capture the dynamic stabilisation of the tetrahedral intermediate. It will give a “green” score for many inactive designs, giving false confidence.

- No free-energy barrier computation: Without a QM/MM activation free-energy estimate (e.g., using EVB or GFN2-xTB), you cannot convincingly claim the design is catalytically competent. These computations require hours per design; you cannot run them in the loop.

2.4 GPU/compute realities

A single RFdiffusion2 backbone-generation run (≥ 100 steps, with substrate conditioning) takes ~ 15 – 20 min on an A100. LigandMPNN + Boltz-2 + PLACER adds another ~ 30 min per design. A co-design loop of, say, 10 iterations would require 8–10 hours of continuous compute — feasible with cloud credits, but fragile. If the loop needs manual curation between steps, you will run out of time.

The polymer-design side is not GPU-accelerated in the same way; you will be stuck with local MD or heuristic selection, creating a computational bottleneck.

2.5 The single biggest technical failure mode

The enzyme-design front-end cannot produce a single plausible carbamate hydrolase candidate within 72 hours that passes both PLACER and an eyeball structural check. You will have a beautiful slide deck, but no actual designed protein structure that a protein engineer would accept as non-obviously broken. The entire co-design illusion falls apart.

3. Scientific Soundness

Even with unlimited time, are the core premises correct?

3.1 Mechanical property targets for the fiber

Your targets — soft-block $T_{g} < -50$ °C, hard-block $T_{g} > 150$ °C, elongation $\geq 400\%$ — are representative of commercial spandex (e.g., Lycra®). They are reasonable but incomplete. A children's onesie also demands:

- Low hysteresis and high elastic recovery ($>95\%$ after 300% strain)

- Abrasion resistance and chlorine resistance
- Compliance with CPSIA lead/phthalate limits, which is not a pure polymer-design problem but affects formulation (additives).
Your generative model does not capture recovery properties or long-term cyclic fatigue; these emerge from the microphase-separated morphology, not just block T_gs.

3.2 Carbamate-accessibility vs. hard-segment crystallinity

Spandex's urethane and urea linkages reside in the hard-segment domains, which are highly hydrogen-bonded and crystalline. Enzymatic attack on a crystalline, water-excluded domain is kinetically crippled. Even if you design an enzyme with an open active site, the polymer must first be swelled or partially amorphised. You have not included a "swelling pre-treatment" step in the design loop. Without it, the in-silico molecular docking is unrealistic — you are docking to an idealised single carbamate chain, not a buried crystalline interface. This is a fundamental disconnect that an astute judge (e.g., a materials chemist from Carter's) will immediately identify.

3.3 Anchoring on UMG-SP2 transition state

UMG-SP2 is a *natural* urethanase that evolved for aromatic carbamates. Its catalytic apparatus (Ser-cis-Lys) is well-characterised by QM/MM [2]. Using that TS as a theozyme anchor is the best available starting point. However, the enzyme's substrate specificity comes from a long binding groove that accommodates the hard-segment diurethane motif. RFdiffusion2 cannot readily create such a complementary groove de novo without extensive hallucination correction. A better (but harder) approach would be to use the entire UMG-SP2 scaffold as a "seed" for hallucination, which is more akin to scaffold-based design than de novo. That would reduce the novelty, but increase feasibility.

3.4 Co-design loop — coherent or hand-wavy?

The loop as described ("fiber's carbamate environment emitted as enzyme target; enzymatic accessibility fed back as fiber constraint") is conceptually sound but computationally ill-defined. What is the quantitative metric for "enzymatic accessibility"? A docking score? A molecular-dynamics-derived conformational exposure of the

scissile bond? You need a differentiable fitness function linking the polymer sequence to the enzyme's predicted k_{cat}/K_M .

In practice, you will resort to a sequential filter: (1) generate polymer candidates, (2) manually inspect hard-segment amorphicity, (3) design enzyme for the most exposed candidate, (4) dock. That is not a tight feedback loop; it is a linear pipeline with hand-offs. The claim of "joint computational co-design" is therefore exaggerated for a 72-hour sprint. A judge familiar with computational design will call this out.

4. Higher-Ceiling Alternatives

Could a different concept on the same battlefield (children's apparel circularity) beat REWEAR-FUSED?

4.1 Alternative A — "Enzyme Cocktail Matched to a Blended Textile" (higher defendability, lower shock)

Design a suite of de novo enzymes that concurrently depolymerise cotton (cellulase), polyester (PETase), and polyurethane (urethanase) from a single process. The computational angle: a multi-objective optimisation that maximises synergy and minimises cross-inhibition. This directly addresses the largest pain point of post-consumer textiles — blends — which is the real Carter's problem (most onesies are cotton/spandex or polyester/spandex).

Why it beats REWEAR-FUSED: It solves the actual mixed-waste stream, not a hypothetical mono-material. The shock factor is high because no one has computationally designed a compatible enzyme cocktail. The commercial story is stronger: "drop your whole garment in, the enzymes sort it out."

Risk: De novo design of three enzymes in 72 hours is even harder; you would likely show a framework and one enzyme.

4.2 Alternative B — "Chemocatalytic Depolymerisation of the Urea Bond" (higher technical floor, less bio-focused)

Instead of an enzyme, co-design an organocatalyst (e.g., a bifunctional thiourea/amine) that selectively cleaves the urea linkage, recovering polyols and amines. The computational loop: a generative model for the catalyst (using SMILES-based RL) paired with the same AI-generative elastane. This avoids the protein-design impossibility while keeping the “co-designed circular pair” narrative.

Why it might beat REWEAR-FUSED: Catalytic small-molecule design is faster and more computationally grounded in DFT, which can be run in 72 hours with cloud resources.

The demo could include a real-time transition-state optimisation.

Why it might not: Carter’s and Cox Cleantech may prize bio-based solutions; the hackathon is sustainability-themed, and enzymatic processes are considered “greener.”

4.3 Alternative C – “Computationally Designed PFAS-free Water-Repellent Finish” (safety, not circularity)

Track is “Making & Remaking,” but children’s apparel has an equally painful problem: durable water repellency without PFAS. You could computationally design a PFAS-free polymer coating that maintains stain resistance. Novelty: de novo design of a functional finish, huge market pull from regulations.

Verdict: Does not address circularity directly; may be off-track.

4.4 Verdict on the best alternative

Alternative A (enzyme cocktail for blends) is the only concept that has a higher ceiling than REWEAR-FUSED while staying on theme. It speaks directly to the real-world barrier that makes Carter’s circularity goals almost impossible (blends). However, it does not carry the elegant “fiber/enzyme co-design” symmetry. If you can merge the two — co-designed polymer/enzyme for a multi-material system — you would be untouchable, but that is far beyond 72 hours. I recommend sticking with REWEAR-FUSED but explicitly calling out the blend-compatibility roadmap in the pitch.

5. Prize-Strategy Critique

What weakens the pitch to Carter’s and Cox Cleantech.

5.1 Carter's-specific blind spots

- Carter's real-world supply chain is 95% cotton/spandex or poly/spandex blends. Your pure elastane-only loop does not solve their current waste stream. They will ask: "How does this separate elastane from cotton in our onesies?" You have no answer in the demo.
- The compliance wrapper (Engine C) is not minor — it is the immediate go/no-go for a children's product. If you demote it to 10% of the demo, you signal that you do not understand the regulatory hurdle that kills most textile-recycling innovations. The wrapper should be a first-class module, not an afterthought.
- The residency thesis — "use \$10K to wet-lab-validate" — is unrealistic. \$10K and 6 weeks is not enough to synthesise a novel elastane, express a de novo enzyme, and measure degradation. You would need to present a staged, milestone-based plan that acknowledges that you will validate a *piece* (e.g., enzyme expression and activity on a model carbamate). Be honest, or the judges will smell naivety.

5.2 Strongest hostile judge question

"You've designed a polymer that doesn't exist and an enzyme that has never been tested, and you claim to close the loop. But a children's onesie is a cotton/spandex blend full of dyes, finishing agents, and unknown contaminants. Even if your enzyme works in a pure system, how do you handle the real garment? And have you checked whether your novel elastane monomer would pass CPSIA migration limits?"

Recommended answer: Acknowledge the blend challenge upfront. Present Engine C as the first screen to sort garments, and propose a pre-treatment step (e.g., mild thermomechanical separation) that you scoped but didn't implement. State that your polymer design incorporates only monomers already approved under CPSIA (e.g., PTMEG, MDI, BDO — which are GRAS for textile use), and that the enzyme design includes no toxic co-factors. Show a quick auto-generated CPSIA-compliance report from your wrapper as proof.

5.3 Most likely reason you lose

1. The enzyme design is judged not credible — no catalytic geometry validated beyond PLACER, no barrier estimate, obvious absence of oxyanion-hole preorganisation.

2. The “joint co-design” is exposed as a linear pipeline with no feedback iteration, undermining the core novelty.
3. A competing team presents a simpler, data-rich, immediately deployable circularity concept (e.g., a digital-twin reverse logistics platform for used onesies) that resonates with Carter’s operations team, while REWEAR-FUSED is dismissed as “deep-tech vapourware.”

6. Threat Ranking & Fixes

Rank	Threat	Severity	Specific Fix for the Hackathon
1	Prior-art scoop on de novo urethanase	Critical	Pre-commit to a theozyme-design methodology that is novel beyond the enzyme itself (e.g., a differentiable active-site hallucination protocol). Even if the enzyme has been designed elsewhere, your <i>method</i> is distinct. Prepare a fallback slide: “If a urethanase appears, we re-base on the newly published structure within the 72 h window to co-optimize the polymer.”

2	Enzyme design output not credible	Critical	Do not claim a high-confidence working enzyme. Instead, present it as a physics-grounded hypothesis with explicit checks: (1) PLACER score, (2) docking pose agreement with QM TS, (3) conservation of oxyanion-hole hydrogen-bond distances. Show failure-mode analysis — this builds trust.
3	Segmented-copolymer generative model intractable	High	Do not attempt a full-blown generative model. Use a template-based enumeration of block sequences around a known spandex formulation, with manual modification guided by accessibility metrics. Frame this as “structure-guided

			AI-assisted redesign,” not pure generation.
4	Ignoring blend reality	High	Dedicate one slide to a down-selection algorithm (Engine C) that identifies and separates elastane-rich fractions using near-infrared spectral signatures (public datasets). It shifts the narrative from “we solved the blend” to “we identified where our solution fits first.”
5	Co-design loop is hand-wavy	Medium	Implement a single explicit feedback step: show the docking score of the designed enzyme on the designed polymer, then adjust the polymer’s accessible-conformation ensemble (e.g., by varying hard-segment length) and re-dock. One loop is enough to prove the concept.

6	Compliance wrapper demotion backfires	Medium	Make Engine C the demo's interactive element — a real-time web-interface where a judge enters a garment barcode and sees a regulatory-decision tree. This is low-effort, high-impact, and shows deployment-readiness.
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7. Final Verdict

REWEAR-FUSED is the highest-ceiling concept I can conceive for this hackathon, provided you execute the demo in a way that foregrounds the audacity of the “molecule/enzyme pair” narrative while being ruthlessly honest about current limits. Its core strength is the symmetry of the closed loop: two computational designs, each feeding the other, to solve a notoriously hard material-circularity bottleneck. No other team is likely to attempt protein design, let alone polymer co-design — that alone gives you the “shouldn’t be possible in 72 hours” shock factor.

However, if your demo does not produce at least one visualisation of a physically plausible enzyme active site with a docked carbamate substrate, and a clear trace of feedback to polymer structure, the concept will collapse under hostile questioning. My recommendation:

1. Pre-build the infrastructure: Have the theozyme, RFdiffusion, and polymer-MD pipelines ready before kickoff. Use the 72 hours to run the loop, not to debug installations.
2. Script the co-design narrative: Open with a manually curated “generation zero” pair, show the incompatibility, then run a refinement step live (or accelerated replay) to demonstrate the loop’s logic.
3. Over-index on the regulatory wrapper demo — it grounds you in Carter’s reality.

4. Prepare the scooped-enzyme counter-narrative: “Our method adapts in real time to the latest published enzyme — here’s how.”

Change nothing about the central idea. But adjust the pitch and the 72-hour scope to survive the kill-shots. If you deliver a demo that presents a plausible carbamate-hydrolysing active site designed by diffusion, a matched polymer candidate, and a working regulatory-check web tool, you will be the team to beat for the grand prize and the Cleantech Residency interview.

References

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All dates and URLs are confirmed as of the simulated search cutoff 31 May 2026.